	TECHNICAL SIGNATORY COMPETENCE DECLARATION AND AUTHORIZATION FORM	
	Document no: ASMF6.2 FORM	
Compiled by:	Tshepo Maponya - Laboratory Supervisor	
Approved by:	Nonhlanhla Msibi - Operations Manager	

PERSONNEL

Candidate name: _____

Position (Technical Signatory): _____

Assessment Date: _____

1. **Technical Knowledge and Theory**
   - The candidate must demonstrate a thorough understanding of the following principles:
   - Standard compliance: Understand the requirements of ISO 1953 for coal sieving.
   - Sample Integrity: Knowledge of the minimum mass requirement (65kg) for statistical representation of top-size particles.
   - Fraction Management: Rationale for "scalping" at 22.4 mm to protect delicate fine mesh sieves
   - Calculation: Fractional Percentage = $(\frac{Mass\ retained}{Total\ mass})\ X\ 100$
2. **Practical Competency Checklist**

Task Item	Competency requirement	Outcome (P/F)
Sample Prep	Correct use of "cone and quarter" method to extract representative sample.	
Drying	Ensure the coal is visibly dry	
Scalping	Proper separation of coarse material and fine fractions	
Coarse analysis	Execution of the 360 °C revolutions, 8 back-and-forth agitations per segment.	
Loading limits	Sample must cover at least ¾ of the total surface area, this to prevent particle breakage and/or restricted movement.	
Fine analysis	Use lids/plastics for dust control and observing the 45-60s settling time.	
Sieving Time	Precise timing of 2 minutes for the fine fraction analysis.	

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3. Data Validation & Quality control

A TS must be able to validate results based on the following criteria:

Mass loss correction: Any difference between the dried initial mass and the sum of recovered fractions must be added back to the "fines" mass (add to the smallest screen).

Recovery limits: Total recovery must lie 99.9≤R≤100.01 (%) to be considered valid.

Corrective action: Identify that dust loss is the most likely source of error if mass balance fails.

Precision: Ensure that major fractions remain within the ~2% absolute repeatability limit as specified by ISO 1953.

4. Results

Sample ID:	
Analysis Date:	
Personnel:	
Initial mass dried (kg):	

Sieve size (mm)	Mass retained (kg)	% Retained
+50mm		
+45mm		
+40mm		
+31.5mm		
+25mm		
+22.4mm		
+12.5mm		
+6.3mm		
+3mm		
+1mm		
-1mm (+mass loss correction due to dust loss)		
Total Recovered		

NB: Empty spaces can be used for other screen sizes

Pass or Fail? : _____

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5. Authorisation

Employee signature:

_____

AUTHORIZING OFFICER

Name: _____

Designation:_____

Date: _____

Authorizing Signature

_____

Declaration: I hereby certify that the candidate has demonstrated the technical proficiency required to review to review, interpret, and sign off on the test results for PSD under AS014IW.

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